

15.1 A 2Tb 4b/Cell 6-Plane 3D-Flash Memory with 37.6Gb/mm² Bit Density and >85MB/s Write Throughput

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Abstract

A 332-WL-layer stacked 2Tb 4b/Cell 6-plane 3D Flash memory achieves an areal density of 37.6Gb/mm². An 8kB-physical and 16kB-logical page architecture realizes a <65μs read latency and a >85MB/s program throughput. Write throughput is enhanced by 4.2% through

voltage-level sensing based on SEN2-LBUS coupling, and an additional 8% by reduced verify operations using 4 check points. A fast SLC burst programming technique reduces program time by ~11%.

The exponential growth of data and expanding AI applications necessitate high-capacity storage solutions with performance that scales with capacity. 3D Flash memory storing 4b/cell offers high density and a lower bit cost, making it well-suited for such applications.

In this work, a 332-WL layer stacked 2Tb 4b/cell 3D-Flash memory achieves an areal density >37Gb/mm². The design incorporates an 8kB physical and 16kB logical page structure, enabling a <65μs read latency and >85MB/s write throughput. Compared to prior 4b/cell designs [1], this work improves density >30% by using a CMOS directly-bonded-to-array (CBA) structure and a 6-plane architecture. 6-plane configuration enhances performance via increased parallelism: supporting asynchronous read operations across all 6 planes. The CBA structure, illustrated in Fig. 15.1.1(bottom right) which shows a peel-back view, also improves power delivery via additional low-resistance metal layers. A novel voltage-level (VL) sensing scheme based on SEN2-LBUS coupling, boosts write throughput and contributes to overall system performance in high-density high-capacity Flash memory systems. To further enhance write throughput, a reduced verify with 4 check points is implemented. Additionally, the fast self-adjusting read (FSAR) mode, which tracks threshold voltage (V_{th}) valley with quick search capability, improves data retention and V_{th} -valley tracking. FSAR achieves faster read performance (t_{READ}) and reduces read retries compared to conventional methods. A fast SLC burst-programming method is introduced to enhance SLC cache-program performance.

This work enables multi-die stacking to support high-capacity storage and leverages CBA technology to overcome limited power pad availability while enhancing power delivery. A new bus-idle sleep (BIS) mode reduces idle currents to near-standby levels, addressing challenges posed by an increased die count. Separate command-address (SCA) protocol is supported for multi-die stack configuration.

Figure 15.1.1(top left) shows the plane architecture of this work; the chip is comprised of 6 physical planes: the 16kB WL is arranged in two groups of 8kB WL, one on each side of the connection region to the stacked WLs. The planes are arranged in a BL direction in a 1x6 configuration. This arrangement enables a ~2.1% area reduction: compared to the 2x3 configuration [1, 5] shown in Fig. 15.1.1(top right). Area reduction is achieved via a smaller pad region that is placed in chip's bottom side and outside of the memory-cell region. A 1x6 configuration is constrained by the number of power pads and vertical power tracks in the x-direction; however, the CBA architecture provides the flexibility to change CMOS wafer process/layers for circuit convenience. Hence, by adding another lower-sheet-resistance metal layer to realize a stronger power mesh to overcome the limitation. In a multi-die scenario, large currents can be injected into package ground, which can result in significant ground bounce that adversely affects the die, performing sense operations. To mitigate ground bounce, we incorporate additional ground pads, as illustrated in Fig. 15.1.1(top left). The main charge-pump circuit, which we have identified as the dominant source of ground bounce, is located on the right side of the chip. We placed additional ground pads on the right side of it to reduce induced ground bounce: even under high-current conditions across the 6-plane configuration. Additionally, a dedicated quiet-ground pad is included to provide an isolated ground connection for circuits that are sensitive to ground noise. These two mitigation strategies enabled the chip to perform asynchronous plane-read operations in a 1x6 configuration.

The need for faster program time (t_{PROG}) is always a benchmark for 3D Flash memory. A lower t_{PROG} is important for 4b/cell flash memories as programming involves verifying 16 states, which can result in a longer t_{PROG} . To reduce SA's x-pitch (to implement CBL), a local clock method was introduced [6]. The sense amplifier is shown in Fig. 15.1.2(top). Utilizing this structure, we propose a new sensing: LBUS coupling sense, which uses SEN2-to-LBUS coupling to directly transfer sense data to a latch without intermediate steps. A typical program verify uses two verify voltage levels per state: a low (V_L) and high (V_H) level [7]. Each verify operation consists of the following four steps: CLK_UP, sensing, CLK_DOWN, and strobe. (1) **CLK_UP** - to obtain a sufficient sense margin, especially for source-biasing sense, we clock up the SEN2 & SEN nodes using LBUS & VLOP; (2) Sensing - next, we sense the data by turning on XXL; (3) **CLK_DOWN** - SEN is clocked down with VLOP; (4) **Strobe** - we then strobe the SEN data onto the DL1 node. Previous overlapping sense

schemes on non-clocking sense amplifiers have been discussed [8]; however, for deeper negative sense schemes it was not applicable due to existing CLK up/down pulses between the two senses. The LBUS coupling-sense scheme allows for an overlapping deep negative sense, which provides a 4.2% t_{PROG} improvement.

For a conventional method, as shown in Fig. 15.1.2(bottom left), step(1) to step(4) are done for VL sensing followed by VH sensing. In the proposed LBUS coupling sense, back-to-back V_L & V_H sensing is done, as shown in Fig. 15.1.2(bottom right). The sense amplifier is shown in Fig. 15.1.2(top). Once V_L sense completes, we transfer SEN2 onto DL1 using SEN2-to-LBUS coupling. For highly conductive cells, T_EN turns on and DL1 flips. For low-conductive cells, T_EN stays off and DL1 doesn't flip. For V_H sensing, we adjust the sense timing for good margin between on cell and off cell. We then clock down the SEN node with VLOP and strobe the data onto DL2 node via A_EN. This proposal needs careful tuning of gate voltages (T_EN) to transfer the information from LBUS to DL1 and to differentiate the on and off cells correctly. In a nutshell, LBUS coupling sense consists of parallel operation (overlap V_L STB and V_H sensing), skipping steps: CLKdown ~ CLKup.

The adequately-reduced-verify mechanism is implemented to optimize program operations in 4b/cell Flash memory: the number of program-verify steps are reduced to maintaining accurate cell-state programming, while improving overall performance and power efficiency. Normally, program-verify operations are performed for all the 15 programmable states. Instead of verifying all the program states, only 4 key states, 4 check points (CP) are verified. Figure 15.1.3(top) shows example CP states and the skipped verify states of the program loop, with the bias condition shown in Fig. 15.1.3(bottom). The program loop number and bias strength are dynamically adjusted based on each check point's verify result. Upper states S12-15 (after CP4 pass) are considered for data retention to optimize bias loop. Compared to all CP, we could get 8% performance gain without losing V_{th} margin.

The proposed FSAR mode addresses the limitations of the conventional valley-tracking-read (VTR) mode by significantly reducing the number of read-retry cycles and by improving read latency (t_{READ}); thereby, enhancing data retention (DR) performance. Compared to the baseline normal mode, the conventional VTR incurs ~7x impact to t_{READ} . In FSAR, quick-search and fine-search modes incur the highest latency (+89%), while quick-search-only and fine-search-only modes offer intermediate performance gains of +35% and +54% respectively. The FSAR technique involves two searching method: the quick-search-mode shown in Fig. 15.1.4(top) and fine-search-mode shown in Fig. 15.1.4(bottom). The supported five read modes are (1) quick-search-only, (2) fine-search-only using the quick-search result, (3) fine-search-only using fine-search result, (4) quick-search and fine-search, and (5) normal read using the quick-search result. Quick-search utilizes bit-count (BC) thresholds and read-level-voltage control adjustments via the following 3-step approach, a 2-point detection read with BC followed by a converting BC to shift value, 2-point-detection read-level adjustments are made using plus and minus shifts from default read level and finally performing page by page read with shift value. Fine-search mode uses a 3-point read method with $\Delta 1$ and $\Delta 2$ comparisons to determine the optimal read levels that involves a 2-step approach. Step 1 proceeds with a 3-point read followed by BC calculation and the storing of the read result in DLs. Step 2 calculates a threshold value and selects the best-read result as per the calculation. Fine-search is to get more accurate read result and the stored fine-search result reflects the next fine-search to save t_{READ} with accurate offset. Quick-search results can be utilized for a fine-search and a normal read without t_{READ} impact. The method supports asynchronous 6-plane read and suspend-read operations for both quick-search and fine-search modes.

To enhance SLC-cache program performance, a fast SLC burst programming method is introduced. The charge pump setup and discharge operations are time consuming for cache-program operations (t_{CPROG}). To reduce charge pump setup and recovery time, a pump set/reset-skip method is proposed. Figure 15.1.5(top) illustrates the pump set/reset-skip scheme. Charge pumps are enabled one time and kept enabled for subsequent program operations as shown in Fig. 15.1.5(bottom). Final discharge is enabled at the end of a burst program sequence. t_{PROG} can save around 11% for each cache program.

As storage density increases, stacking more dies as shown in Fig. 15.1.6(top left) becomes essential. In multi-die architectures, the idle current from unselected dies is approaching active-current levels. To minimize idle current, the data path must be fully shut down to imitate standby mode. An additional feature, BIS mode, is introduced to reduce the idle current from 10's of mA to 100's of μ A for each die when enabled. The controller sends a LUNSEL command followed by the address of the target die. All of the stacked die with common CEnx that do not match the address issued will go into BIS mode: where an internal chip-enable signal, CEni, as shown in Fig. 15.1.6(top right), is disabled; thereby, disabling the complete data path and voltage generators, behaving as if the die was in standby mode. To exit BIS mode, the controller only drives CEnx to logic-high, which resets all gating, as shown in Fig. 15.1.6 (bottom).

The die micrograph is shown in Fig. 15.1.7(bottom) and a comparison of this work with prior 4b/cell and 3b/cell 3D Flash memory is given in Fig. 15.1.7(top).

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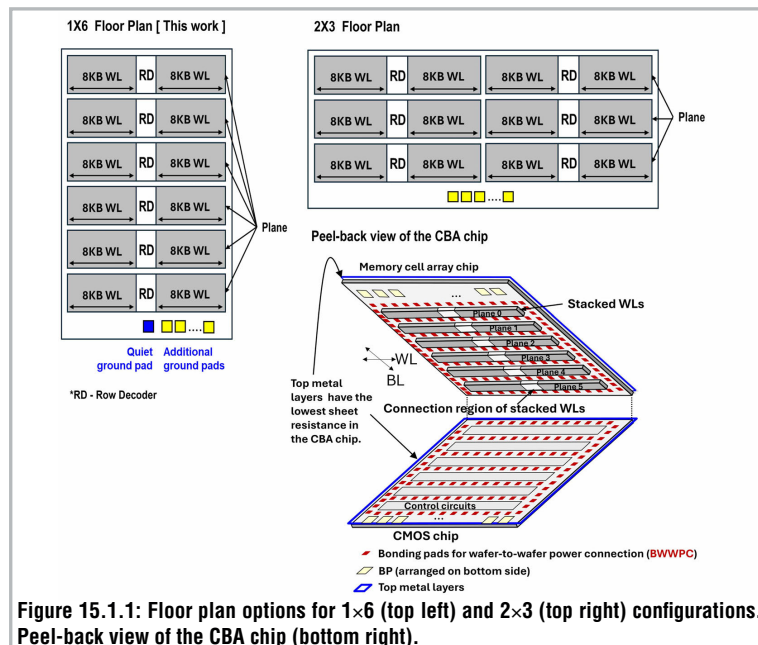


Figure 15.1.1: Floor plan options for 1x6 (top left) and 2x3 (top right) configurations. Peel-back view of the CBA chip (bottom right).

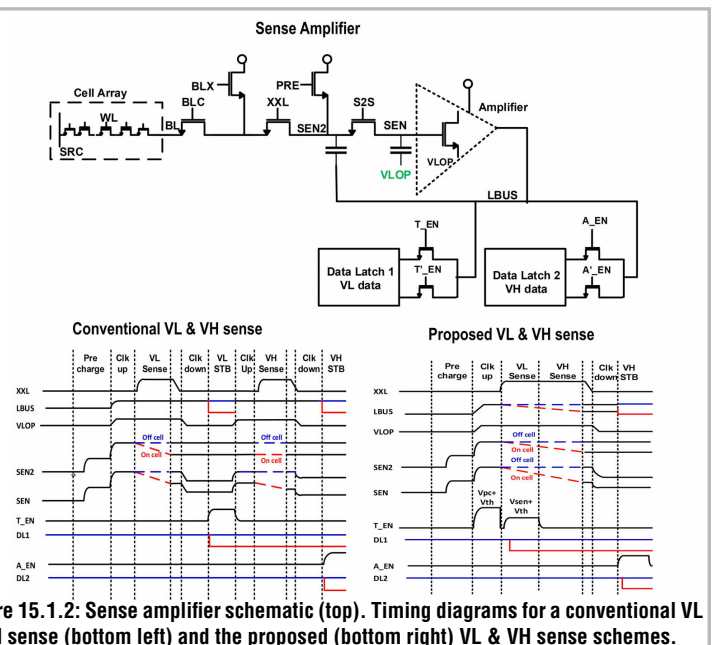


Figure 15.1.2: Sense amplifier schematic (top). Timing diagrams for a conventional VL & VH sense (bottom left) and the proposed (bottom right) VL & VH sense schemes.

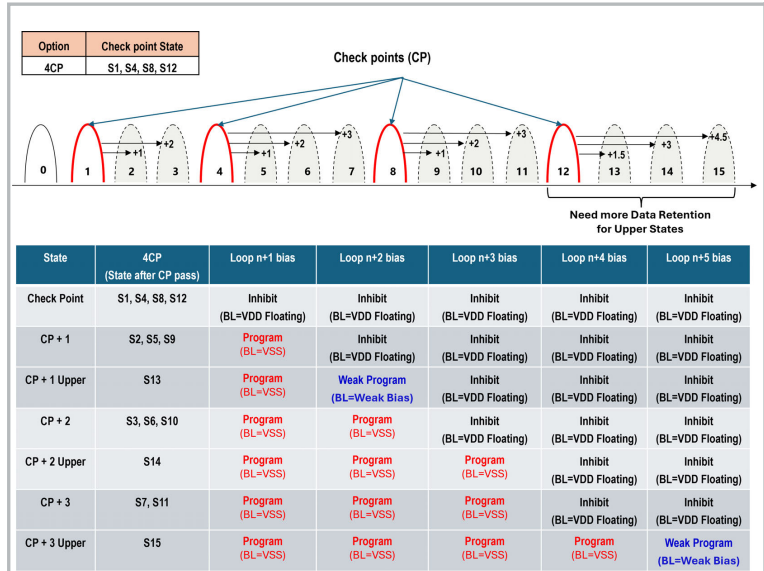


Figure 15.1.3: Adequately-reduced program verify with 4 check points. Example check-points states (top) and program loop with bias condition (bottom).

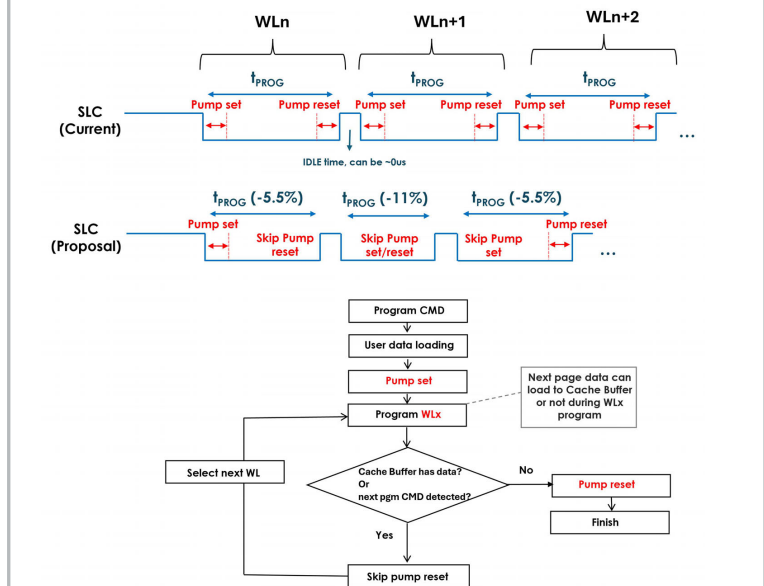


Figure 15.1.5: Fast SLC burst program with pump set/reset-skip method (top), and flow diagram for enabling the charge pump (bottom).

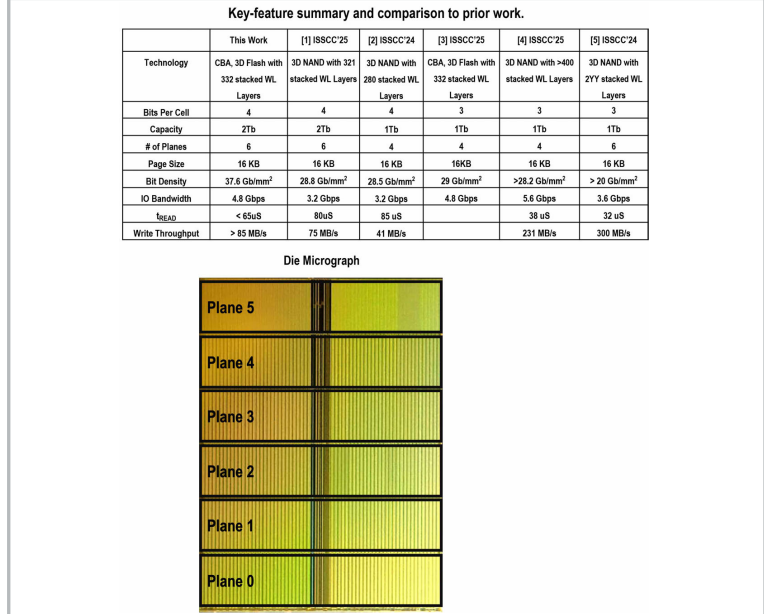


Figure 15.1.7: Key-feature comparison table and die micrograph.

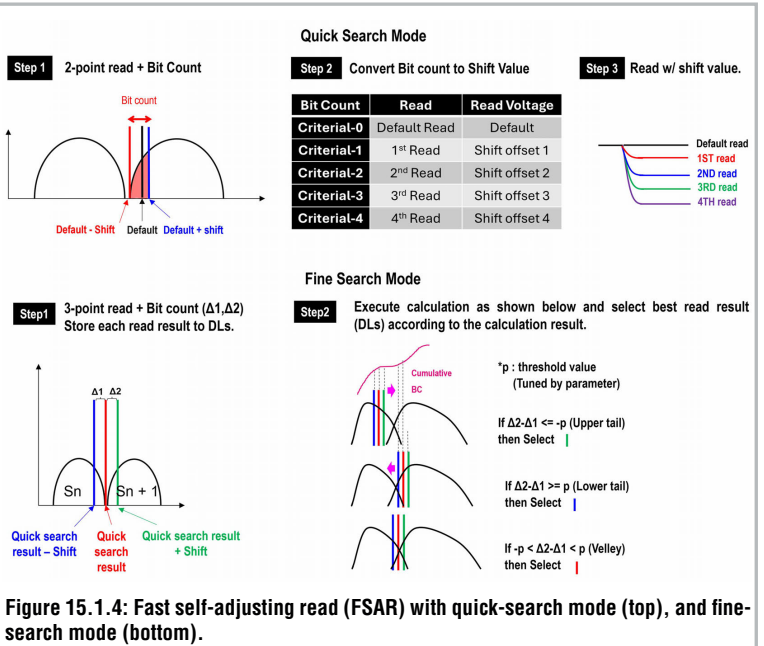


Figure 15.1.4: Fast self-adjusting read (FSAR) with quick-search mode (top), and fine-search mode (bottom).

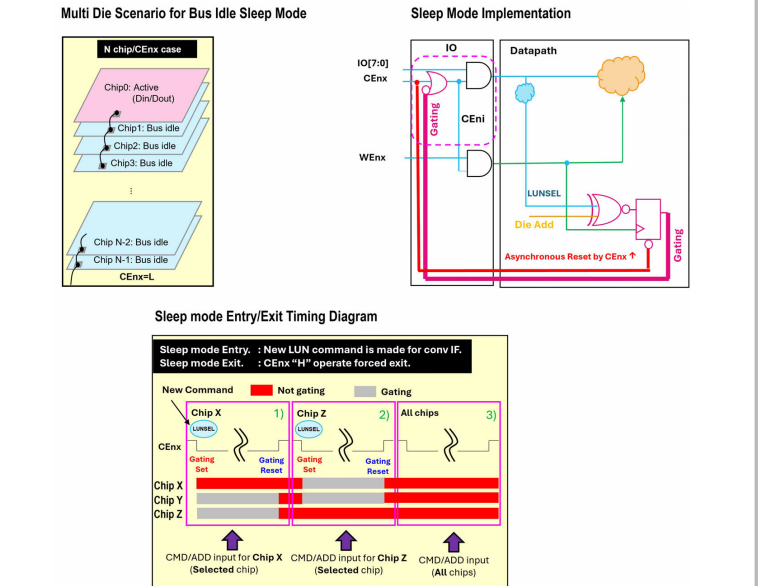


Figure 15.1.6: Bus-idle-sleep mode for multi-die scenario (top-left), sleep-mode implementation (top right), and entry/exit timing diagram (bottom).